

## Smart Metrics

Design metrics supporting log/audit/monitoring, directly computable from project's output

| Metric Name                      | Formula/Definition                                                            | Data Source                                  | Update Cadence                             | Intended Use/Req Link                                             | 800-55 Term                 |
|----------------------------------|-------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|-----------------------------|
| <b>Source Coverage Ratio</b>     | # of unique PQC sources used in artifacts / total sources cited               | docs/source_count.csv                        | Weekly (after each literature search step) | 03.02.01 (info baseline), mitigates "scope bias" risk             | Measure selection           |
| <b>Reference Freshness Score</b> | Avg(year of sources) or % of sources published within last 3 years            | docs/source_log.csv                          | Weekly                                     | 03.02.03 (content update), counters outdated info risk            | Data quality/prioritization |
| <b>Peer Review Validation %</b>  | # of research findings cross-checked by a secondary reviewer / total findings | docs/peerreview_log.txt                      | After each experimental run/review cycle   | 03.05.01 (accuracy/relevance review), mitigates error propagation | Data collection             |
| <b>Privacy Incident Rate</b>     | # of detected test runs with leakage / total test runs                        | src/paramleak_check.py, logs/test_output.log | Per test batch or weekly                   | 03.13.02 (confidentiality), addresses privacy/leakage risk        | Measure prioritization      |
| <b>De-ID Enforcement Ratio</b>   | # of datasets successfully de-identified / total datasets analyzed            | docs/data_deid.log                           | After each new dataset intake              | 03.13.01 (data separation), controls re-identification risk       | Usefulness                  |

|                                          |                                                                                            |                                                          |                                                   |                                                                             |                         |
|------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------|-------------------------|
| <b>Secure<br/>Storage<br/>Compliance</b> | # of datasets<br>stored per<br>encryption/co<br>mpliance<br>policy / total<br>PQC datasets | src/storage_policy_check.py, log<br>s/storage_output.log | Weekly<br>or after<br>any<br>new<br>stored<br>set | 03.08.02<br>(storage<br>protection),<br>mitigates<br>unsafe<br>sharing risk | Assurance/s<br>election |
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